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Keystroke dynamics to detect Parkinson's disease

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Abstract

This project investigated whether ensemble machine learning classifiers can reliably detect Parkinson’s disease from keystroke dynamics under subject-wise evaluation. Prior work on the Tappy keystroke dataset reported accuracies approaching 100% under restrictive conditions that had not been independently tested on a broader sample. A pipeline engineered 51 features per participant and evaluated Random Forest and XGBoost under nested subject-wise cross-validation on 131 participants (98 Parkinson’s, 33 controls). Neither model reliably separated the classes: both showed near-zero specificity and performed at or below the majority-class baseline on threshold-dependent metrics. Feature ablation showed that central tendency features carried more usable signal than variability features, and a fold-specific stability analysis identified recurring direction-specific timing features but no globally stable subset. Stable-subset retraining improved some ranking metrics and reduced dimensionality but did not resolve the control-class failure; SMOTE shifted the decision boundary at substantial cost to overall performance. The high accuracies reported in prior Tappy-based work depended on specific experimental conditions and did not generalise under broader subject-wise evaluation. The project contributes a reproducible benchmark showing that weak, unstable class-discriminative signal, rather than model complexity, is the main limitation on this dataset.

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List of Acronyms

AUC	Area Under the Curve
BCS	British Computer Society
CNN	Convolutional Neural Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
CSV	Comma-Separated Values
CV	Coefficient of Variation
DaTscan	Dopamine Transporter Scan
F1	Harmonic Mean of Precision and Recall
FN	False Negative
FP	False Positive
IQR	Interquartile Range
KNN	K-Nearest Neighbours
LDA	Linear Discriminant Analysis
LSTM	Long Short-Term Memory
MAD	Median Absolute Deviation
MIT-CSXPD	Massachusetts Institute of Technology - Clinical Symptoms of Parkinson's Disease
PD	Parkinson's Disease
PR-AUC	Precision-Recall Area Under the Curve
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
UPDRS-III	Unified Parkinson's Disease Rating Scale Part III
XGB	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background

Parkinson’s disease (PD) is the second most common neurodegenerative disorder worldwide, affecting an estimated ten million people globally and placing a substantial burden on individuals, caregivers, and healthcare systems (Kalia & Lang 2015). It is characterised by the progressive degeneration of dopaminergic neurons in the substantia nigra, producing motor symptoms such as tremor, rigidity, bradykinesia, and postural instability. A major clinical limitation is that these symptoms become clearly observable only after substantial and irreversible neurological damage has already occurred (Kalia & Lang 2015). Earlier, non-invasive detection therefore remains an important clinical challenge.

Current diagnosis relies primarily on specialist clinical assessment, sometimes supported by confirmatory procedures such as Dopamine Transporter scan (DaTscan) or skin biopsies. These methods are costly, invasive and not equally accessible across healthcare settings, particularly outside major urban centres (Sun et al. 2024). Clinical assessment also depends on judgement of the examining clinician, which introduces subjectivity into the diagnostic process. A reliable behavioural marker that could be captured passively and objectively would therefore have practical value as a supplementary screening aid.

One promising source of such a marker is fine motor behaviour during typing. Subtle motor irregularities associated with PD may emerge before overt symptoms become clinically obvious, including changes in movement speed, timing, and coordination (Adams 2017). Keystroke dynamics captures these behaviours through the timing of key presses and releases during ordinary computer use. Unlike imaging or laboratory based assessments, keystroke data can be collected unobtrusively using standard hardware in everyday settings. Prior work has shown that typing-derived features can retain discriminative value even under unsupervised, at-home conditions, suggesting that the underlying motor signal is not confined to tightly controlled laboratory environments (Arroyo-Gallego et al. 2018).

Machine learning classifiers trained on keystroke timing features have been reported to distinguish Parkinson’s disease patients from healthy controls with reported accuracies typically ranging between 70% and 100%, depending on dataset, feature representation, and evaluation procedures (Adams 2017, Demir et al. 2023). However, these headline results do not by themselves establish that a classifier has learned a clinically meaningful

signal. In practice, a screening or monitoring system would only ever be applied to individuals whose data the model had not previously encountered. The central issue is therefore not simply whether a model can separate classes within a dataset, but whether it can generalise to unseen participants under defensible evaluation conditions.

The issue is particularly important for the Tappy dataset, which has become a widely used benchmark in keystroke-based Parkinson’s research. Adams (2017) reported near-perfect classification performance on a heavily filtered subset of 53 participants after dimensionality reduction. Demir et al. (2023) reported similarly high results using a much larger feature space and multiple sub-datasets, but did not clearly document whether subject-level separation was enforced during validation. These studies demonstrate that strong results can be obtained on Tappy under specific experimental conditions, but they leave open a different and practically significant question: how do models behave when those conditions are tightened around the requirement of generalisation to unseen users?

This study addressed that gap by evaluating Random Forest (RF) and XGBoost under identical preprocessing and subject-wise validation conditions on the Tappy dataset. No study identified during this review directly tested whether the high performance reported in earlier Tappy-based work holds when restrictive participant selection is removed and models are evaluated strictly on unseen individuals. Rather than treating model comparison as a search for the highest score alone, the project examined what each model learned when required to classify participants not present during training, whether that learned signal remained stable across different folds, and what the findings imply for the plausibility of keystroke-based Parkinson’s screening in real-world use.

The full project proposal, including the original aims, objectives, and project plan, is provided in Appendix A.

1.2 Aims and Objectives

This project aimed to investigate whether Random Forest and XGBoost can reliably classify Parkinson’s disease from keystroke dynamics when evaluated on unseen participants, and to examine what signal, if any, supports their predictions under those conditions.

The comparison was conducted under identical preprocessing, feature engineering, and subject-wise validation conditions using the Tappy keystroke dataset, ensuring that any observed differences between the models could be attributed as far as possible to learning behaviour rather than experimental inconsistency. Subject-wise GroupKFold cross-validation ensured that no participant appeared in both training and test sets within any fold, making the reported results relevant to the harder and more realistic task of classifying new individuals.

Objective 1 - Data acquisition, preprocessing, and feature engineering

The Tappy keystroke dataset was acquired and consolidated into a unified processing pipeline. Raw keystroke logs were parsed and standardised, with systematic handling of missing or inconsistent entries, and participant identifiers were preserved to support subject-wise splitting. Timing features derived from hold time, flight time, and latency were then aggregated at participant level to produce one feature vector per user.

Deliverable: A cleaned, feature-engineered dataset with one row per participant, supported by summary statistics and visualisations, ready for model training.

Objective 2 - Model development and optimisation

Random Forest and XGBoost classifiers were implemented and trained under identical preprocessing and validation conditions. Hyperparameters were optimised using cross-validated grid search, and all training procedures were documented to support reproducibility.

Deliverable: Two trained and optimised classifiers with documented configurations, training logs, and reproducible pipelines.

Objective 3 - Evaluation and analysis

Both models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC, with results interpreted against a majority-class baseline. Beyond aggregate performance, the analysis examined model behaviour under subject-wise evaluation through feature ablation, feature stability analysis, and a supplementary SMOTE experiment. These additions extended the original plan in response to the behaviour observed in the primary comparison.

Deliverable: A comparative evaluation report including performance tables, confusion matrices, feature importance and stability analyses, and a critical discussion of what the models learned under subject-wise evaluation.

1.3 Methodology

This project adopted a quantitative, experimental research methodology structured around the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Chapman et al. 1999), which provides defined stages for data understanding, preparation, modelling and evaluation while permitting iteration between them. The framework suited a project centred on systematic experimentation, where the feature ablation and stability analyses emerged from interpretation of the primary results rather than being fixed in advance.

The dataset used was the publicly available Tappy keystroke dataset (Goldberger et al. 2000), selected because it is the principal benchmark in keystroke-based Parkinson’s research and was collected under unsupervised home-typing conditions relevant to real-world screening. Its self-reported diagnoses and limited timing measurements are constraints considered when interpreting the results.

Implementation used Python 3.12, with scikit-learn (Pedregosa et al. 2011) providing the Random Forest implementation, GroupKFold cross-validation, GridSearchCV and the evaluation metrics, and the XGBoost library (Chen & Guestrin 2016) providing the gradient-boosted classifier. Random Forest (Breiman 2001) and XGBoost (Chen & Guestrin 2016) were selected for a controlled pairwise comparison within the same ensemble tree family, the rationale is developed in Section 2.5. Evaluation used accuracy, precision, recall, F1, ROC-AUC and PR-AUC, with PR-AUC included because it is more informative than ROC-AUC under class imbalance (Saito & Rehmsmeier 2015). The full

requirements and pipeline design are presented in Chapter 3, with implementation in Chapter 4.

1.4 Report Roadmap

The remainder of this report follows the progression of the project from research context to implementation and evaluation.

Chapter 2 reviews the literature on keystroke-based Parkinson’s detection and identifies the methodological issues most relevant to this study, particularly those affecting the interpretation of model performance and generalisability. Chapter 3 translates these issues into the requirements and design of the experimental pipeline. Chapter 4 describes how that design was implemented in code, including the data preparation, feature engineering, model training, and testing procedures.

Chapter 5 presents the experimental results from the primary model comparison and the extended analyses, and evaluates them in relation to the project objectives and the problem of subject-wise generalisation. Chapter 6 considers the legal, social, ethical, and professional implications of a system with the performance characteristics observed in this study. Chapter 7 concludes the report by reflecting on the key findings, limitations, lessons learned, and directions for future work.

Chapter 2

Literature Review

2.1 Introduction

This chapter examines the existing research relevant to keystroke-based Parkinson’s disease detection, with the review organised around four themes:

- The clinical basis for using keystroke dynamics as a biomarker.
- What previous classification studies actually demonstrate about model performance and what their experimental designs allow the reader to conclude.
- Which methodological choices determine whether reported results can be interpreted as evidence of generalisation to unseen individuals rather than performance under narrow experimental conditions.
- Why ensemble tree methods were selected over alternative classification approaches for the present study.

The literature is not reviewed here as a simple sequence of reported accuracies. High classification scores alone do not establish that a model has learned a clinically meaningful or transferable signal. For this project, the more important issue is whether the experimental design supports conclusions about how a classifier would behave when applied to a new participant whose data was not available during training. This chapter therefore examines prior work in terms of dataset characteristics, feature representation, validation design, interpretability, and ecological validity. These themes directly inform the design decisions made in the subsequent chapters.

2.2 Keystroke Dynamics as a Behavioural Marker of Parkinson’s Disease

The motor symptoms of Parkinson’s disease introduced in Section 1.1, which include bradykinesia, rigidity, tremor, and postural instability (Kalia & Lang 2015), typically become observable only after substantial neurological damage has already occurred. This has motivated interest in behavioural markers capable of detecting earlier and more objective motor change.

Keystroke dynamics offers a mechanism for capturing these fine motor changes passively during everyday computer use. Typing requires repeated, time-sensitive finger movements, and each keystroke produces a series of measurable temporal events. Table 2.1 defines the three core timing features captured in keystroke data and their relationship to Parkinson’s disease motor symptoms.

Feature	Definition	Effect of PD
Hold time	The duration a key is pressed, measured from key press to key release.	Bradykinesia causes slower key presses and longer hold times.
Latency	The interval between pressing one key and pressing the next.	Rigidity and bradykinesia increase latency and introduce greater variability across successive keystrokes.
Flight Time	The interval between releasing one key and pressing the next.	Tremor and impaired coordination introduce asymmetry between left and right hand transitions.

Table 2.1: Keystroke timing features and their relationship to Parkinson’s disease motor symptoms

These measurements are relevant because Parkinsonian motor impairment affects movement speed, regularity, and coordination. Adams (2017) argued that these subtle disturbances may appear in everyday typing before they become obvious in routine clinical examination, making keystroke timing a plausible candidate digital biomarker.

The appeal of this approach is not only clinical but also practical. Unlike imaging procedures or laboratory-based motor tests, typing data can be collected using standard computer hardware without specialist equipment, clinical visits, or active patient effort (Sun et al. 2024). This positions keystroke dynamics as a scalable screening tool that could operate continuously in the background of a person’s daily computer use, detecting changes over time rather than relying on a single clinical snapshot. Arroyo-Gallego et al. (2018) validated this concept by demonstrating that keystroke features collected during unsupervised typing at home could distinguish early-stage Parkinson’s patients from controls, confirming that the motor signal is detectable outside controlled laboratory conditions. This finding is important because it suggests that keystroke-based detection may remain feasible under realistic conditions, while also indicating that ecological validity introduces additional noise that can reduce classification performance.

The datasets used in this field reflect this trade-off between control and realism. The NeuroQWERTY MIT-CSXPD dataset, hosted on PhysioNet, contains keystroke logs from 85 participants collected under clinically supervised conditions on a standardised laptop, with confirmed diagnoses and UPDRS-III motor severity scores (Goldberger et al. 2000). Its controlled collection environment provides stronger diagnostic labels and more standardised data, but captures behaviour under artificial conditions that may not reflect everyday typing.

In contrast, the Tappy dataset, created by Adams, was collected in unsupervised home settings over extended periods, giving it greater ecological relevance but also greater variability. Participants used their own computers and typed in naturalistic contexts

rather than under a standardised protocol. For the present project, this makes Tappy especially relevant, since any realistic screening or monitoring tool would need to operate under similarly uncontrolled conditions. However, Tappy imposes clear limitations such as the Parkinson’s label being self-reported rather than clinically verified, and the data records only a limited set of keystroke timing measures. These features of the dataset affect both the strength of the available signal and the confidence with which classification results can be interpreted.

2.3 Prior Classification Studies and What They Demonstrate

The most prominent study on the Tappy dataset is that of Adams (2017), who applied an eight-model ensemble consisting of Naive Bayes, Decision Tree, Random Forest, Neural Networks, Support Vector Machine, Logistic Regression, K-nearest Neighbours, and Gradient Boosting. Before training, Adams filtered the dataset to only include 53 participants meeting strict criteria of a minimum of 2,000 keystrokes, mild Parkinson’s disease severity only, and no Levodopa medication, then applied linear discriminant analysis (LDA) to reduce the dimensionality of a 27-feature space (9 hold-time features and 18 latency features, each comprising mean, standard deviation, skewness, kurtosis, and asymmetry measures). Under these conditions the ensemble achieved classification accuracies approaching 100% using 10-fold cross-validation at the participant level. While this result is frequently cited as evidence that keystroke dynamics can reliably detect Parkinson’s disease, several aspects of the experimental design limit the strength of this conclusion. The heavy participant filtering reduced the sample to roughly a quarter of the full dataset and, as an additional consequence, inverted the class composition: Adams’s analysis subset contained 20 PD and 33 control participants, a near-balanced ratio that does not exist in the broader dataset. This raises the question of whether the reported performance reflects features of Parkinson’s disease itself or features of the specific subset selected, with the favourable class composition further contributing to the apparent ease of separation. The use of LDA further complicates interpretation, as it projects features into a lower-dimensional space optimised for class separability, meaning the classifier operates on transformed features rather than the original keystroke measurements. No subsequent study identified during this review has attempted to replicate Adams’s results under less restrictive conditions, leaving it unclear whether the reported performance generalises beyond the specific pipeline he constructed.

Demir et al. (2023) revisited the Tappy dataset using a substantially different methodology. Rather than applying a single feature set to a single model, Demir et al. (2023) extracted 378 features from the keystroke data and organised them into 14 sub-datasets, each representing a different grouping of feature types. They then applied Random Forest, Support Vector Machine, and K-Nearest Neighbours classifiers individually, as well as through feature ensemble and model ensemble methods that combine predictions from sub-datasets. Their reported accuracies ranged from 81% to 100% depending on the sub-dataset and ensemble configuration. The breadth of this approach makes direct comparisons with Adams difficult, where Adams used 53 participants with LDA and fixed feature set, Demir et al. (2023) used the full participant pool with 378 features and combinatorial ensemble methods. The headline accuracy range is therefore not directly

comparable to Adams’s result, as two studies operated under fundamentally different experimental conditions. Critically, Demir et al. (2023) did not document their validation procedure clearly enough to verify subject-level separation, making it unclear whether the reported accuracies reflect the model’s ability to generalise to unseen individuals or its ability to recognise patterns within individuals already represented in the training data.

Outside the Tappy dataset, Tripathi et al. (2022) applied Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to the NeuroQWERTY MIT-CSXPD dataset, achieving accuracies between 88% and 95%. Their use of deep learning represents a methodological departure from the ensemble and classical machine learning approaches applied to Tappy, and the NeuroQWERTY dataset provides clinically verified diagnoses and UPDRS-III motor severity scores that Tappy lacks. However, NeuroQWERTY was collected under standardised laboratory conditions with a single laptop model, making it a controlled but less ecologically valid testbed compared to Tappy’s uncontrolled home environment. The higher accuracies reported by Tripathi et al. (2022) may therefore reflect both the benefits of deep learning architectures and the reduced noise in a clinically supervised dataset, rather than a generalisable advantage of CNNs or LSTMs over tree-based classifiers. This study therefore broadens the methodological landscape of keystroke-based detection but does not resolve the specific question of how models generalise on other datasets such as Tappy under subject-wise evaluation.

Arroyo-Gallego et al. (2018) addressed the ecological validity question directly by comparing classification performance between clinically supervised typing and unsupervised home typing using the NeuroQWERTY framework. They reported an AUC of 0.83 under supervised conditions and 0.76 in the home environment, demonstrating that motor signal captured by keystroke dynamics degrades but remains detectable when participants type freely on their own devices. This finding is directly relevant to interpreting results on the Tappy dataset, where all data was collected under unsupervised conditions. The performance ceiling on Tappy is expected to be lower than on clinically controlled datasets, and moderate classification accuracy should be understood as reflecting the ecological conditions of the data rather than the inadequacy of the model.

A systematic review of technology-based Parkinson’s diagnosis by Alfalahi et al. (2022), covering gait, speech, handwriting and keystroke data, found significant heterogeneity in datasets, feature extraction methods, and analysis approaches. They concluded that the lack of standardised features and inconsistent analysis methods remain a fundamental challenge for cross-study comparison in this field, particularly for keystroke approaches where the number of available studies is small relative to gait and speech.

Study	Dataset	N	Features	Model(s)	Validation	Best Acc.
Adams (2017)	Tappy	53	27 + LDA	8 model ensemble	10-fold CV (subject)	~100%
Demir et al. (2023)	Tappy	227	378	RF,SVM,KNN + ensembles	10-fold (subject separation unclear)	81-100%
Tripathi et al. (2022)	NeuroQWERTY	85	CNN/LSTM input	CNN,LSTM	Not documented	88-95%
Arroyo-Gallego et al. (2018)	NeuroQWERTY	52	nQi features	Logistic Regression	Supervised vs home	AUC 0.76-0.83
This project	Tappy	131	51	RF, XGBoost	5-fold GroupKFold	71.8%

Table 2.2: Comparison of keystroke-based Parkinson’s disease classification studies

The Table 2.2 illustrates three critical observations. First, no two studies used the same

experimental configuration, confirming the concern raised by Alfalahi et al. (2022) that cross-study comparison in this field is unreliable. Second, the only prior study on Tappy that documented subject-level validation is Adams, whose sample was heavily filtered and reduced to 53 participants. Third, no prior work identified during this review has compared Random Forest and XGBoost under identical, transparent conditions on the Tappy dataset, nor has any study conducted a feature importance analysis identifying which keystroke characteristics contribute most to classification, an analysis Adams himself identified as future work. The present project addresses these gaps by using a broader participant pool than Adams without restrictive filtering, documents the validation protocol explicitly using GroupKFold to enforce subject-wise separation, and provides both a controlled model comparison and feature importance analysis under valid experimental conditions.

A further methodological challenge visible in the table is the class imbalance inherent in the Tappy dataset. After applying the minimum keystroke threshold of 2,000, the dataset used in this project contains 98 participants with Parkinson’s disease and 33 controls, producing an approximate 3:1 class ratio.

The implications of this imbalance for classifier behaviour and metric choice are addressed in Section 2.4. The relevant gap for the present chapter is that no study identified during this review evaluated keystroke-based Parkinson’s classification on the imbalanced full sample, nor applied rebalancing techniques such as Synthetic Minority Over-sampling (Chawla et al. 2002) on this dataset. Adams (2017)’s filtering pipeline incidentally produced a near-balanced subset of 20 PD and 33 controls, but the imbalance was avoided through participant exclusion rather than addressed as a methodological issue, and the question of how classifiers behave on the imbalanced broader sample remains open. The present project addresses this gap directly.

2.4 Evaluation Challenges and Methodological Limitations

The studies reviewed in the previous section reported classification accuracies ranging from 70% to 100%, but the variation in experimental conditions makes it difficult to determine how much of this range reflects genuine model capability and how much is an artefact of methodological differences. Four recurring issues limit the interpretation of reported performance in this field and directly shaped the design of the present project:

- Participant filtering and the limits of narrow samples.
- Validation design and what it determines about generalisation.
- Class imbalance as a source of misleading metrics.
- Interpretability and the stability of learned signal.

The first is participant filtering and the limits of narrow samples. Restrictive selection criteria can make classification appear easier by removing the variability that exists in the broader dataset. This does not make the result invalid, but it limits what the result can be taken to show. A model that achieves high accuracy on a carefully filtered subset has demonstrated that it works under those specific conditions, not that it would perform equally well on a more representative sample. For a study concerned with real-world screening plausibility, broader inclusion is often more informative even when it produces lower scores. The present project retained only the 2,000 keystroke minimum filter, which serves a data quality purpose rather than a clinical selection purpose, and applied it to the full participant pool to produce a sample of 131 users that is more representative of the broader Tappy population.

The second issue is validation design and what it determines about generalisation. In classification tasks where data is grouped by participant, the method used to split training and test sets determines what the reported accuracies actually measures. If keystrokes or feature vectors from the same participant appear in both the training and test sets, the model can achieve high accuracy by learning individual typing patterns rather than patterns associated with Parkinson’s disease. Kaufman et al. (2012) demonstrated that such leakage produces overly optimistic performance estimates by allowing information from the evaluation set to influence training. In keystroke analysis, this problem is especially relevant because typing behaviour contains person-specific regularities unrelated to disease status. Where validation protocols are not clearly documented, it is impossible to determine whether reported accuracies reflect genuine generalisation to unseen individuals or are inflated by within-subject leakage. The present project used GroupK-Fold cross-validation with the user identifier as the grouping variable, ensuring that no participant’s data appeared in both training and test sets of any fold.

The third issue is class imbalance as a source of misleading metrics. When one class substantially outnumbers the other, standard accuracy metrics can mask poor discriminative ability. As He & Garcia (2009) showed, classifiers trained on imbalanced data tend to favour the majority class, producing high recall for that class at the expense of the minority class. In the context of Parkinson’s detection on the Tappy dataset, this means

a classifier could achieve approximately 75% accuracy by predicting Parkinson’s for every participant without learning any meaningful signal from the keystroke features. Saito & Rehmsmeier (2015) argued that precision-recall analysis is more informative than ROC analysis under such conditions, since it better reflects the trade-off between positive prediction quality and coverage of the positive class. For the present study, this means that performance interpretation must consider whether a model is genuinely discriminating between classes or merely exploiting prevalence structure. The present project adopted PR-AUC as the primary supplementary metric and included a SMOTE experiment to test whether rebalancing the minority class altered classification behaviour.

The fourth issue is interpretability and the stability of learned signals. Even when a classifier produces strong performance, the result reveals little about what behavioural information the model is using unless some account of feature contribution is provided. Feature importance methods offer one route into this problem by identifying which variables contribute most to classification decisions (Louppe et al. 2013). However, a single importance ranking from a single training run is not sufficient evidence that the identified features carry a genuine and stable signal. If feature importance changes substantially across different training samples, the apparent signal may be an artefact of particular fold compositions rather than a consistent property of the data. The question is therefore not only which features appear important, but whether they remain important across different subsets of unseen users. A model that performs well for unstable or unclear reasons offers limited scientific and practical value. The present project addressed this by computing feature importance across all cross-validation folds and conducting a stability analysis to identify which features were consistently ranked as important regardless of fold composition.

These four issues are not independent. Restrictive filtering can mask the effects of class imbalance by producing a more balanced subset. Undocumented validation can inflate results that would otherwise reveal weak generalisation. High reported accuracy without feature importance analysis provides no basis for determining whether the classifier learned clinically plausible signal or dataset-specific noise. The experimental design of the present project was constructed to address each of these issues simultaneously: broader participant inclusion, documented subject-wise validation, imbalance-aware metrics and rebalancing, and feature importance with stability testing.

2.5 Model Selection Rationale

The choice of Random Forest and XGBoost was driven by the specific aims of this project rather than by a general preference for popular models. The project required two classifiers that could operate effectively on a relatively small participant-level tabular dataset, support a controlled comparison under identical conditions, and allow direct inspection of the signal used for classification.

Random Forest (Breiman 2001) and XGBoost (Chen & Guestrin 2016) satisfy these requirements while differing in an analytically useful way. Both are tree-based ensemble methods, making them comparable within the same broad model family. RF uses bagging, constructing multiple independent trees and combining their outputs by majority

vote, whereas XGBoost builds trees sequentially with each tree trained to correct the residual errors of the previous one through gradient boosting. This shared architectural foundation means that any observed performance difference can be attributed to the learning strategy rather than to fundamental differences in model structure. Critically, both models provide native feature importance scores through the accumulation of impurity reduction across all trees, enabling direct inspection of which features each model relies on without requiring additional post-hoc methods such as permutation importance or SHapley Additive exPlanations (SHAP) analysis. This made them well suited to a pairwise comparison centred not simply on which model scores higher, but on how two related ensemble strategies behave under the same subject-wise evaluation conditions and what signal each learns to exploit.

There is also precedent for using Random Forest and XGBoost in small biomedical tabular problems. Hong et al. (2022) compared the two methods for severity prediction in a cohort of 63 COVID-19 patients and showed that both can perform effectively under limited sample conditions. This does not make that problem directly comparable to keystroke-based Parkinson’s detection, but it supports the suitability of these models for a participant-level biomedical dataset of modest size.

Alternative models were considered but were less suitable for the present research question. Deep learning architecture, particularly CNNs and LSTMs, have demonstrated strong performance on keystroke data in other contexts (Tripathi et al. 2022), but they require substantially larger training sets to learn meaningful representations without overfitting. With only 131 participants, the available training data is insufficient to support the number of learnable parameters in even a modestly sized neural network. Deep learning approaches also function as black-box models, making it difficult to extract interpretable information about which features drive classification, which conflicts with the interpretability objective of this project.

Support Vector Machines (SVMs) have been applied to keystroke-based Parkinson’s detection by both Adams (2017) and Demir et al. (2023). SVMs are effective in high-dimensional spaces and can handle small sample sizes. However, they do not provide built-in feature importance measures without additional techniques such as permutation importance or recursive feature elimination, and their performance is sensitive to kernel selection and hyperparameter tuning in ways that complicate fair comparison, weakening their fit for a project where inspecting the learned signal is a core objective. More importantly, the research question focuses specifically on comparing bagging and boosting strategies within the ensemble tree family, placing SVMs outside the scope of the controlled comparison.

Logistic Regression offers simplicity and full interpretability through its coefficient weights, and has been used in keystroke-based detection by Arroyo-Gallego et al. (2018). However, logistic regression assumes a linear relationship between features and the log-odds of the outcome, which may not hold for the complex interactions between keystroke timing features and Parkinson’s disease status. Tree-based methods make no such assumption, as they partition the feature space through recursive splitting and can capture non-linear relationships and feature interactions without requiring explicit specification.

The decision to compare two models rather than a larger set was deliberate. A broader multi-model sweep may increase the number of reported results, but it can dilute analytical clarity. A controlled pairwise comparison under identical conditions produces cleaner evidence about the effect of the learning strategy itself than a configuration sweep across multiple unrelated model families, which is the approach taken in both Adams (2017) and Demir et al. (2023).

2.6 Conclusion

The literature reviewed in this chapter supports keystroke dynamics as a plausible behavioural marker of Parkinson’s disease, but it also establishes that reported classification performance cannot be interpreted without examining the conditions under which it was obtained. Keystroke timing features capture measurable motor impairment associated with Parkinson’s disease, and this signal persists even under unsupervised typing conditions, although with reduced discriminative power compared to clinically controlled environments (Arroyo-Gallego et al. 2018). However, the classification accuracies reported in prior work on the Tappy dataset range from 70% to 100%, and this variation is attributable to differences in participant filtering, feature representation, and validation protocol rather than to meaningful differences in model capability. Studies that reported the highest accuracies either applied restrictive filters that reduced the dataset to a narrow, homogeneous subset (Adams 2017) or did not document their validation protocol clearly enough to verify subject-level separation (Demir et al. 2023). No study identified during this review conducted a feature importance analysis on the Tappy dataset, despite Adams (2017) identifying this as an area for future work, and no study identified during this review addressed the 3:1 class imbalance that is present in all configurations of the dataset.

These findings informed four design decisions for the project. The minimum 2,000 keystroke filter was retained for data quality, but the severity and medication filters used by Adams (2017) were removed to produce a broader and more representative sample. Subject-wise GroupKFold cross-validation was adopted to ensure that classification performance reflects generalisation to unseen individuals rather than recognition of individuals already present in training data. Random Forest and XGBoost were selected as a controlled pairwise comparison isolating the effect of bagging versus boosting within the ensemble tree family, with both models providing native feature importance scores. A supplementary SMOTE experiment was included to test whether rebalancing the minority class improves classification under the corrected validation conditions. The following chapter documents the requirements and design of the pipeline constructed to implement these decisions.

Chapter 3

Requirements Analysis and Design

3.1 Introduction

This chapter documents the requirements that the experimental pipeline needed to satisfy and the design decisions made to meet them. Because this project produced a data preprocessing and classification pipeline rather than a user-facing application, the requirements were derived from the methodological gaps identified in the literature review and from the constraints imposed by the dataset, rather than from a user story or stakeholder consultation. The design section presents the architecture of the pipeline and justifies each component with reference to the literature and the project objectives.

3.2 Requirements Analysis

3.2.1 Requirements Derivation

The requirements for this project were derived from three sources. First, the literature review identified specific methodological weaknesses in prior work on the Tappy dataset: restrictive participant filtering, undocumented validation protocols, and the absence of feature importance analysis. Any pipeline designed to address these weaknesses needed to avoid introducing the same problem. Second, the Tappy dataset itself imposed constraints on what was technically feasible, including the format of the raw data files, the self-reported nature of the diagnoses, and the class imbalance between Parkinson’s disease and control participants. Third, the research question required a controlled comparison between two specific models under identical conditions, which placed requirements on how data was preprocessed, how hyperparameters were tuned, and how results were evaluated.

3.2.2 Assumptions and Constraints

Several assumptions were made during the development of the pipeline. The self-reported Parkinson’s disease diagnoses in the Tappy dataset were treated as the reference labels for the classification, as no clinical verification was available. This introduces label noise that may reduce the achievable separation between classes, a limitation that is considered when interpreting the results. Participants who provided fewer than 2,000 keystrokes were assumed to have insufficient data for reliable feature extraction, following the threshold established by Adams (2017). Three timing features recorded per keystroke (hold time, latency, and flight time) were assumed to capture meaningful motor information, based on the clinical rationale established in Section 2.2 linking keystroke timing to bradykinesia, rigidity, and tremor.

The primary constraint was dataset size. After applying the minimum keystroke filter, 131 participants remained, of whom 98 had PD and 33 were controls. This 3:1 class imbalance constrained model training, evaluation metric selection, and the feasibility of techniques such as deep learning that require larger sample sizes. A further constraint was computational: hyperparameter tuning using nested cross-validation with GroupKFold was computationally expensive, which influenced the decision to limit the grid search space to a manageable number of configurations rather than performing exhaustive search.

3.2.3 Functional Requirements

The functional requirements were prioritised using Must have, Should have, Could have, and Won’t have (MoSCoW) to distinguish between components essential to answering the research question and those that would strengthen the analysis without being strictly necessary.

Must have: The pipeline must load, parse, and merge the raw Tappy keystroke log files and metadata files into a single consolidated dataset. The pipeline must compute per-user aggregate features from the raw keystroke data, including statistical summaries of hold time, latency, and flight time, and direction-specific transition features. The pipeline must train Random Forest and XGBoost classifiers under identical preprocessing and validation conditions. The pipeline must implement subject-wise GroupKFold cross-validation ensuring no participant appears in both training and test sets within any fold. The pipeline must report accuracy, precision, recall, F1-score, PR-AUC, and ROC-AUC with mean and standard deviation across folds.

Should Have: The pipeline should perform hyperparameter tuning using GridSearchCV with GroupKFold inner cross-validation. It should compute feature importance scores across folds, report confusion matrices for error analysis, and include a majority-class baseline so that model performance can be interpreted against the class imbalance present in the dataset.

Could have: The pipeline could include feature ablation, feature stability analysis and a SMOTE experiment to test how feature grouping, feature consistency, and synthetic oversampling affect classification behaviour under subject-wise validation.

Won’t have: The pipeline will not implement deep learning models, as the dataset size

does not support them. The pipeline will not include a user interface, as the output is an experimental evaluation rather than a deployable tool.

3.2.4 Non-functional Requirements

Reproducibility: All random seeds must be fixed and all hyperparameter configurations documented so that results can be independently replicated. This was prioritised as the highest non-functional requirement because the central contribution of the project is a methodological correction. If the pipeline itself is not reproducible, the argument loses credibility.

Transparency: Every design decision in the pipeline must be traceable to either a finding from the literature review or a constraint imposed by the dataset. No black-box preprocessing steps or undocumented transformations should be applied.

Modularity: The pipeline should be separated into distinct scripts corresponding to CRISP-DM stages (data cleaning, feature engineering, model training, and supplementary analysis) so that each stage can be run and verified independently.

3.3 Design

3.3.1 Pipeline Architecture

The pipeline was designed as a modular sequence of scripts aligned with the CRISP-DM framework adopted in Section 1.3. Intermediate datasets and experiment outputs were saved after each major stage, allowing the cleaned keystroke data, participant-level feature matrix, model-training results, ablation results, stability outputs, and SMOTE outputs to be inspected independently. This structure supported transparency and reproducibility by making it possible to verify each stage without rerunning the full pipeline.

Stage 1 - Data cleaning (`Tappy_clean.py`): loads the raw Tappy keystroke log files and metadata files, parses the tab-delimited keystroke records, merges them with participant metadata on user identifier, handles missing and malformed entries, and outputs a single consolidated CSV containing all valid keystroke records with associated participant information including Parkinson’s disease status.

Stage 2 - Feature Engineering (`Feature_engineering.py`): reads the cleaned keystroke data, applies the minimum 2,000 keystroke filter, and aggregates the per-keystroke records into a single vector per participant. The feature set consists of 51 engineered timing features derived from hold time, latency, and flight time. These features are organised into two groups: 20 central tendency features capturing average typing behaviour, and 31 variability features capturing inconsistency in typing timing. The output is a 131-row participant-level feature matrix, with one row per user, used as the input for model training and supplementary analysis.

Stage 3 - Model Training and Evaluation (`Model_training.py`): reads the participant-level feature matrix and evaluates RF and XGBoost using nested GroupKFold cross-validation. StandardScaler is applied inside a scikit-learn pipeline, while GridSearchCV performs

hyperparameter tuning using three-fold inner GroupKFold and final performance is estimated using five-fold outer GroupKFold. The script records per-fold metrics, confusion matrices, selected hyperparameters, and feature-importance scores. Supplementary scripts apply the same subject-wise structure to the ablation, stability, and SMOTE analyses, with SMOTE applied only to the training fold before model fitting.

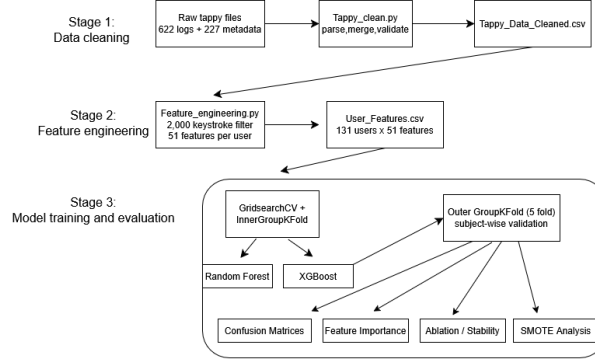


Figure 3.1: Pipeline architecture showing the CRISP-DM-based preprocessing, feature engineering, nested GroupKFold model evaluation, and supplementary analysis stages

3.3.2 Design Decisions and Rationale

Four design decisions shaped the pipeline, each motivated by a specific issue identified in Section 2.4 and each carrying a trade-off worth recording.

User-level feature aggregation

Aggregating per-keystroke records into one feature vector per participant was a deliberate design defence against the within-subject leakage risk identified in Section 2.4. If individual keystrokes were used as separate training instances, records from the same participant could appear in both training and test folds, allowing the model to learn person-specific typing patterns rather than disease-related signal and producing inflated performance estimates. With each participant contributing one row, this division cannot occur, and GroupKFold provides an explicit safeguard at the participant level. The trade-off is reduced temporal granularity: aggregation discards information about how typing behaviour changes across sessions or over time. A session-level representation could capture this signal but would require stricter grouping controls.

Removal of severity and medication filters

The filtering critique in Section 2.3 motivated retaining only the 2,000-keystroke quality filter, producing a 131-participant sample more representative of the broader Tappy population. The trade-off is sample heterogeneity: Adams (2017)’s filters produced a clinically uniform cohort in which class separation may have been clearer. The present project evaluates a more heterogeneous and therefore more challenging classification problem, which contributes to lower observed accuracy.

Inclusion of direction-specific features

The clinical rationale in Section 2.2 linking tremor and impaired coordination to asymmetry between hand transitions motivated the LL, LR, RL, and RR transition features. Their inclusion also maintains comparability with Adams (2017), who used direction-specific features.

F1-score as the GridSearchCV tuning metric

The class imbalance issue identified in Section 2.4 made accuracy an inappropriate tuning objective, since a model predicting Parkinson’s for every participant would score around 75% without learning any signal. F1 penalises models that achieve high recall by predicting the majority class indiscriminately. The trade-off is that F1 weights precision and recall equally, which may not reflect a clinical screening priority where higher recall at the cost of precision could be preferred.

3.4 Conclusions

The requirements analysis identified a set of functional and non-functional specifications driven by the methodological gaps in prior work and the constraints of the Tappy dataset. The pipeline was designed as a three-stage modular architecture aligned with CRISP-DM, with each design decision traceable to a finding from the literature review. The key design choices, user-level feature aggregation, removal of restrictive clinical filters, subject-wise cross-validation, and F1-based hyperparameter tuning were all motivated by the need to produce valid, reproducible results on the more demanding task of generalising to unseen individuals. These decisions involved trade-offs between granularity and validation integrity, sample homogeneity and representativeness, and metric sensitivity and clinical relevance. The evaluation chapter revisits whether these trade-offs were justified in light of the experimental results. The following chapter documents how this design was implemented.

Chapter 4

Implementation

4.1 Introduction

This chapter documents the implementation of the experimental pipeline designed in Chapter 3. The pipeline consisted of six Python scripts, each corresponding to data cleaning, feature engineering, model training and evaluation, feature ablation, feature stability analysis, and SMOTE analysis. The chapter is organised by pipeline stage, focusing on the implementation decisions that were central to producing valid and reproducible results.

4.2 Code Repository Outline

The pipeline was implemented entirely in Python 3.12 using the PyCharm integrated development environment. The code was developed by the author with the aid of online tutorials, YouTube instructional content, and AI-assisted debugging during development. The core logic, experimental design, and all implementation decisions were developed independently. External resources were used to resolve specific technical issues such as library syntax, debugging unexpected errors, and understanding API behaviour for scikit-learn and XGBoost. No code was copied from existing implementations of the keystroke-based Parkinson’s detection or adopted from prior work on the Tappy dataset. The pipeline consisted of six scripts executed sequentially, with intermediate datasets and experimental outputs saved after each major stage.

`tappy_clean.py` cleaned and consolidated the raw keystroke logs and participant meta-data. `Feature_engineering.py` generated the participant-level feature matrix. `Model_training.py` produced the primary RF and XGBoost evaluation results. `Ablation_experiment.py` ran the feature-group ablation experiment. `Feature_stability.py` produced the fold-wise feature-importance and stability outputs. `SMOTE_experiment.py` evaluated the over-sampling condition.

The following external libraries were used: `pandas`, `openpyxl`, and `numpy` for data manipulation, `scikit-learn` (Pedregosa et al. 2011) for Random Forest, StandardScaler, GroupKFold, GridSearchCV, and all classification metrics, `xgboost` (Chen & Guestrin 2016) for the XGBoost classifier, and `scipy` for the median absolute deviation calculation used in feature engineering. All libraries were installed via pip.

4.3 Data Cleaning and Consolidation

The Tappy dataset presented a non-trivial data engineering challenge. The raw data consisted of 622 separate keystroke log files and 227 separate user metadata files, each stored as a tab-delimited text file with no standardised header format. The keystroke log files contained one row per keypress event with fields for user identifier, date, timestamp, hand (left or right), hold time, direction of the key translation (LL,LR,RL,RR), latency between consecutive key presses, and flight time between consecutive key release and key press events. The metadata files contained participant identifiers linked to self-reported Parkinson’s disease status, disease severity, medication usage, and demographic information.

The cleaning script (`tappy_clean.py`) parsed all 622 keystroke files, extracted the relevant fields, and merged them with the metadata on the shared user identifier. Rows with missing or malformed timing values were removed rather than imputed, as the volume of keystroke data per participant (typically tens of thousands of records) means that removing individual malformed rows had negligible impact on the aggregate statistics computed during feature engineering. The output was a single CSV file containing all valid keystroke records with their associated participant metadata, preserving the user identifier as the grouping variable for downstream processing.

An early implementation decision concerned the handling of the severity and medication filters used by Adams (2017). As discussed in Section 2.3, Adams applied three filters that reduced the dataset from over 200 participants to 53. The filtering logic was implemented in the pipeline but commented out after initial testing, as described in Section 3.3.2. This allowed the pipeline to be re-run with filters enabled for comparison purposes if needed, while the primary analysis used the full 131 participant sample. The commented-out code is preserved in the submitted source to demonstrate that this was a deliberate methodological choice rather than an oversight, and to allow the pipeline to be rerun under Adams’s original conditions for verification if required.

4.4 Feature Engineering

The feature engineering stage (`Feature_engineering.py`) transformed the cleaned per-keystroke records into a single feature vector per participant. This aggregation removed the record-level structure that could otherwise allow multiple records from the same participant to be split across training and test folds.

The final feature set consisted of 51 participant-level timing features derived from the original Tappy timing variables: hold time, latency, and flight time. In addition to central tendency features describing average typing behaviour, the implementation included variability-specific metrics motivated by the clinical literature linking PD to motor inconsistency. These variability features included coefficient of variation (standard deviation divided by the mean), median absolute deviation, the 90th–10th percentile spread and a burstiness index defined in Equation 4.1:

$$\frac{\sigma - \mu}{\sigma + \mu} \tag{4.1}$$

computed for each of the three timing variables. The burstiness index captures whether a participant’s keystroke timing is characterised by bursts of irregular activity or by relatively even spacing. A value near -1 indicates regular periodic timing, near 0 indicates random timing, and positive values indicate bursty, irregular timing.

The following code snippet illustrates the computation of the variability features for a single timing variable within the per-user aggregation function:

```
mean_val = values.mean()
std_val = values.std()
if mean_val != 0:
    features[f"{col}_cv"] = std_val / mean_val
else:
    features[f"{col}_cv"] = 0

features[f"{col}_mad"] = (
    median_abs_deviation(values, nan_policy="omit")
    if len(values) > 0 else 0
)

features[f"{col}_q90_q10"] = (
    values.quantile(0.9) - values.quantile(0.1)
    if len(values) > 0 else 0
)

if (std_val + mean_val) != 0:
    features[f"{col}_burstiness"] = (std_val - mean_val) / (std_val + mean_val)
else:
    features[f"{col}_burstiness"] = 0
```

The guard clauses were introduced to handle zero denominators, as a small subset of participants exhibited constant timing values for certain keystroke transitions which produced division by zero errors. Rather than imputing these cases, the features were set to zero, which correctly represents the absence of variability.

The 51 features were organised into two conceptual groups that formed the basis of the ablation experiment. Central tendency features (20 features) captured average typing behaviour through global and direction-specific means and medians for each timing variable and per-hand hold time means. Variability features (31 features) captured motor inconsistency through standard deviation, skewness, kurtosis, IQR, coefficient of variation, MAD, quantile spread, and burstiness for each timing variable, plus per-hand standard deviations, direction-specific standard deviations, and a hand asymmetry ratio. This separation was motivated by the clinical question of whether Parkinson’s disease is better detected through how slowly a person types or how inconsistently they type.

4.5 Model Training and Evaluation

The model training script (`Model_training.py`) implemented a fully nested cross-validation design. An earlier version of the pipeline tuned hyperparameters using `GridSearchCV` with three-fold `GroupKFold` on the full scaled feature matrix, then applied the selected hyperparameters across the five-fold outer `GroupKFold` evaluation loop. While participant-level aggregation prevented within-subject leakage in the standard sense, this design allowed information from each outer test fold to influence hyperparameter selection through the tuning step. Because the project’s central argument concerns the dependence of prior Tappy results on validation choices, holding the implementation to the same standard required full nesting. The pipeline was therefore restructured so that scaling and tuning are performed exclusively within each outer training fold, and all reported results in this paper were generated with this corrected design.

The corrected design has three components. `StandardScaler` and the classifier are wrapped inside a scikit-learn `Pipeline`, so scaling is fitted only on training data within each fold. Hyperparameter tuning is performed independently within each outer training fold using a three-fold inner `GroupKFold`, so the search itself never sees the outer test fold. The best estimator selected by each inner search is evaluated once on the untouched outer test fold. Because tuning runs independently per outer fold, the optimal configuration is allowed to differ across folds, which is what distinguishes nested cross-validation from the simpler pre-tuned variant. F1 was retained as the inner optimisation objective for the reasons given in Section 3.3.2.

The following code snippet illustrates the inner-fold tuning structure as applied inside each iteration of the outer loop:

```
rf_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", RandomForestClassifier(random_state=42))
])

rf_grid = GridSearchCV(
    estimator=rf_pipeline,
    param_grid=rf_param_grid,
    cv=GroupKFold(n_splits=3),
    scoring="f1",
    n_jobs=-1,
    refit=True
)

rf_grid.fit(X_train, y_train, groups=groups_train)
```

The `groups=groups_train` argument ensures the inner three-fold split respects participant boundaries. Without it, the inner tuning would silently revert to standard splitting, compromising the nested design at the inner level even though the outer level remained group-aware.

Both models were tuned using predefined hyperparameter grids. The Random Forest grid covered `n_estimators` (100, 200, 300), `max_depth` (5, 10, 15, None), and `min_samples_split`

(2, 5, 10), with `class_weight="balanced"` applied to address the 3:1 class imbalance. The XGBoost grid covered `n_estimators` (100, 200, 300), `max_depth` (3, 5, 7), and `learning_rate` (0.01, 0.05, 0.1), with `scale_pos_weight=1`. The asymmetric treatment reflects preliminary experiments in which increasing `scale_pos_weight` produced marginal recall gains at substantial precision cost, lowering overall F1. Per-fold metrics, confusion matrices, and feature importances were recorded for every outer fold, along with the best hyperparameter configuration selected by each inner search.

A majority class baseline was computed for each fold by predicting the training set’s most frequent label for all test instances. This baseline ranged from 69.2% to 81.5% accuracy and F1 from 0.818 to 0.898 across folds, providing a reference point against which model performance could be judged. Models that did not exceed this baseline were therefore interpreted cautiously, as their aggregate performance may reflect the class distribution rather than reliable class-discriminative signal.

The implementation recorded accuracy, precision, recall, F1, PR-AUC and ROC-AUC for every fold and configuration, alongside confusion matrices to expose class-specific failure patterns that aggregate metrics can obscure. The rationale for prioritising F1 and PR-AUC under the 3:1 imbalance is given in Sections 2.4 and 3.3.2.

Inner-tuning F1 was recorded alongside outer-test F1 for each fold as a direct check on whether the inner search converged on configurations that transferred to unseen participants. These values were retained in the structured experiment outputs to verify that model selection occurred fully within the nested design.

4.6 Feature Ablation Experiment

The ablation experiment (`Ablation_experiment.py`) was implemented as a stand-alone script to maintain modularity and to allow independent verification. The purpose of the experiment was to decompose classification performance by feature family, isolating whether the discriminative signal in keystroke data comes primarily from central tendency features capturing average typing speed, variability features capturing motor inconsistency, or the combination of both. Feature ablation was chosen over alternative feature selection methods such as recursive feature elimination or wrapper-based selection because it preserves the interpretability of the grouping. Recursive feature elimination removes individual features based on model-specific rankings, which can obscure whether the underlying signal is related to average behaviour or to inconsistency. Ablation by conceptual group directly answers a clinically meaningful question of whether the motor signature of PD is better captured by how slowly a person types or by how irregularly they type, which individual feature removal is unable to address. The experiment ran three conditions: central tendency features only (20 features), variability features only (31 features), and combined (51 features). Under the same nested cross-validation structure used in the primary analysis, hyperparameters were re-tuned independently for each condition to ensure fair comparison, since the optimal configuration for 20 features may differ from that for 51 features.

The feature grouping was defined programmatically rather than manually to reduce the

risk of classification errors. Each of the three timing variables contributed its mean and median to the central tendency group, and its standard deviation, skewness, kurtosis, IQR, coefficient of variation, MAD, quantile spread, and burstiness index to the variability group. Direction-specific means were assigned to central tendency, direction-specific standard deviations to variability. The script verified that no feature appeared in both groups and printed any features not covered by either group, providing an automated consistency check.

4.7 Feature Stability Analysis

The feature stability script (`Feature_stability.py`) addressed whether the features identified as important by the models were consistent across training samples or were artefacts of particular fold compositions. Standard feature importance reporting averages a single ranking across folds, which obscures whether the same features are important in every fold or whether the average is dominated by a few folds in which a feature happened to rank highly. The stability analysis examined importance at the fold level, treating consistency across folds as evidence of genuine signal. The analysis was restricted to the 20 central tendency features, since the ablation in Section 4.6 had established that these carried the dominant discriminative signal.

To prevent outer-test-fold information from leaking into feature selection, stability detection was nested entirely inside each outer training fold. The script performs three operations within each outer fold, an inner stability scan to identify a fold-specific stable subset, retraining of both models on that subset, and evaluation on the outer test fold. The stable subset is allowed to differ across outer folds, which is a consequence of the design rather than a flaw, because selecting features globally would leak and fixing them in advance would defeat the purpose.

Within each outer training fold, the stability scan was implemented as follows. The outer training data was split using a three-fold inner `GroupKFold`. For each of the three inner folds, the model was tuned on the inner training set using the same nested procedure as the primary analysis (a further `GroupKFold`-based `GridSearchCV` over the same hyperparameter grids), and the top-10 features by importance were recorded from the resulting best estimator. A feature was classified as stable if it appeared in the top 10 of all three inner folds. If no features met this strict 3/3 threshold, the threshold was relaxed to 2/3 to ensure that each outer fold produced a non-empty stable subset. The 3/3 threshold was preferred as the default because it provides the strongest evidence of consistency, while the 2/3 fallback ensured the script could not silently fail in folds where the inner partitions were unusually variable.

Selection was performed independently for Random Forest and XGBoost, producing two stable subsets per outer fold combined by union. Intersection was rejected because, on a feature space of only 20 features, requiring agreement between both models would have produced subsets too small for meaningful retraining, and would have biased the result toward features that bagging and boosting happen to rank similarly rather than features that are genuinely stable.

Once the union stable subset was determined for an outer fold, both Random Forest and XGBoost were retrained on the outer training data restricted to that subset, with

hyperparameters re-tuned through the same inner-GroupKFold GridSearchCV used in Section 4.5. The retrained models were evaluated once on the untouched outer test fold, with per-fold metrics, confusion matrices, and the selected feature list recorded. The aggregate results across the five outer folds were saved to `Feature_stability.xlsx`, with per-fold stable feature lists retained for inspection in Chapter 5. The mean number of features in the stable subset across the five outer folds is reported in Section 5.2.4.

This design converts feature importance from a descriptive output into a testable claim. If the stable subset produces comparable or better performance than the full feature set, the remaining features are adding noise rather than signal, and because the subset was selected without ever seeing the outer test fold, this comparison is leakage-free. The per-fold stable lists themselves are reported in Chapter 5.

4.8 SMOTE Experiment

The SMOTE experiment (`SMOTE_experiment.py`) tested whether synthetic oversampling of the minority class altered classification behaviour under the subject-wise validation conditions used throughout the project. SMOTE (Chawla et al. 2002) was chosen over random undersampling because the control class contained only 33 participants. Random undersampling would have reduced the PD class to a comparably small size, discarding the majority of available training data and producing an unstable classifier. SMOTE preserves all available data while generating synthetic minority instances through interpolation between existing controls, which is more appropriate for a small dataset where information loss carries a higher cost than the introduction of synthetic points.

The script replicated the full nested cross-validation structure of the primary analysis, with one modification: SMOTE was incorporated into an `imblearn` Pipeline alongside `StandardScaler` and the classifier, so that within each outer training fold, synthetic control instances were generated until both classes were equal in size. Critically, because SMOTE was a step in the pipeline, it was fitted only on the training data within each inner-fold split during hyperparameter tuning and only on training data within each outer fold during evaluation. Hyperparameters were re-tuned within each outer fold using the same inner-GroupKFold GridSearchCV procedure as the primary analysis, meaning the SMOTE comparison isolates the effect of rebalancing under otherwise identical nested validation conditions.

4.9 Testing

Testing focused on verifying that each pipeline stage produced correct outputs and that the experimental design was implemented as intended. Four categories were applied: covering data integrity, cross-validation integrity, ablation consistency, and reproducibility.

4.9.1 Data Integrity Testing

Row counts, column counts, class counts, and key timing-value ranges were checked at each pipeline stage. After feature engineering, the matrix contained 131 rows and 51 feature columns plus `user_id`, `pd_label`, and `keystroke_count`, with class distribution of 98

PD and 33 control participants. No NaN values were present in the final feature matrix, consistent with the guard clauses used in the variability-feature computation.

4.9.2 Cross-Validation Integrity Testing

The GroupKFold implementation was verified by printing training and test sizes, PD counts, and control counts for each fold. This confirmed that no fold had an empty minority class and that fold sizes were approximately equal across the five splits. The non-overlap of participants between training and test sets was guaranteed by the scikit-learn GroupKFold implementation, with the per-fold printout providing additional confirmation that fold sizes and class distributions were as expected.

4.9.3 Ablation Consistency Testing

The ablation script included automated verification that the central tendency and variability feature groups had no overlap and that their union covered all features in the dataset. This was printed at the start of each run and confirmed manually before the experimental results were trusted. The fold compositions across all three ablation conditions were identical, since GroupKFold with the same groups and number of splits produces deterministic fold assignments, ensuring that performance differences between conditions were attributable to the feature set rather than to differences in which participant appeared in which fold.

4.9.4 Reproducibility Testing

All random seeds were fixed (`random_state=42` for all classifiers) and the pipeline was run end-to-end twice with the saved outputs compared for consistency. This confirmed that the results presented in Chapter 5 are deterministically reproducible from the submitted code.

4.10 Conclusion

The implementation produced a six-script pipeline that progressed from raw Tappy data files to a complete set of experimental results, including primary model comparison, feature ablation, feature stability analysis, and a SMOTE experiment. Each script was designed to be independently verifiable through its saved intermediate datasets or structured experiment outputs, supporting the modularity and transparency requirements established in Section 3.2.4. The testing procedures confirmed that the pipeline produced correct outputs at each stage and that the experimental design, particularly the subject-wise cross-validation and the feature grouping for ablation, was implemented as intended. The most significant implementation challenges were the non-trivial data engineering required to consolidate the raw Tappy files and the careful handling of edge cases in the variability feature computation. The following chapter presents the results produced by this pipeline and evaluates them against the project’s objectives and the existing literature.

Chapter 5

Results and Evaluation

5.1 Introduction

This chapter presents the experimental results produced by the pipeline described in Chapter 4 and evaluates them against the project objectives, the majority-class baseline, and the prior work reviewed in Chapter 2. The results are organised around four analyses: the primary RF and XGBoost comparison, the feature ablation experiment, the feature-importance and stability analysis, and the supplementary SMOTE experiment. The evaluation section then interprets these findings in relation to subject-wise generalisation, class imbalance, feature stability, and the practical limits of keystroke-based Parkinson’s screening.

5.2 Results

The results are presented in the order in which the experiments were conducted, all generated using the nested subject-wise validation described in Section 4.5.

5.2.1 Primary Model Comparison

The primary experiment compared Random Forest and XGBoost on the full 51-feature set across the 131 participants. Table 5.1 reports mean performance across the five outer folds. The majority-class baseline achieved 0.748 accuracy and 0.855 F1 by predicting the training-fold majority class; neither trained model exceeded this baseline on either metric.

Metric	Random Forest	XGBoost
Accuracy	0.680 (± 0.039)	0.718 (± 0.018)
Precision	0.736 (± 0.044)	0.746 (± 0.047)
Recall	0.904 (± 0.105)	0.954 (± 0.079)
F1	0.807 (± 0.031)	0.834 (± 0.014)
PR-AUC	0.708 (± 0.071)	0.692 (± 0.117)
ROC-AUC	0.371 (± 0.052)	0.320 (± 0.103)

Table 5.1: Primary classification results on the 51 feature set using five-fold GroupKFold

As shown in Table 5.1, XGBoost achieved marginally higher accuracy and F1, with lower variance on both (accuracy std 0.018 versus 0.039; F1 std 0.014 versus 0.031). Random Forest produced the higher PR-AUC (0.708 versus 0.692), suggesting slightly stronger ranking performance despite weaker threshold-based classification (Saito & Rehmsmeier 2015). The ROC-AUC values (0.371 for Random Forest, 0.320 for XGBoost) are below 0.5, indicating that on the held-out folds the models systematically rank actual Parkinson’s participants below actual controls. This is consistent with the weak and unstable class-discriminative signal explored in the rest of this chapter, rather than evidence of a calibrated but inverted classifier.

The confusion matrices confirm this limitation: both models correctly identified only 1 of 33 control participants, showing that their high recall mainly reflected a tendency to predict Parkinson’s rather than reliable discrimination.

	Predicted Control	Predicted PD
RF — Actual Control	1	32
RF — Actual PD	10	88
XGB — Actual Control	1	32
XGB — Actual PD	5	93

Table 5.2: Aggregated confusion matrices for the primary results (51 features)

As shown in Table 5.2, specificity for both models was 3.0%. Random Forest misclassified 10 PD participants as controls. XGBoost misclassified 5.

5.2.2 Feature Ablation

The ablation experiment tested whether the discriminative signal in keystroke data comes primarily from central tendency features capturing average typing speeds, variability features capturing motor inconsistency, or the combination of both. Three conditions were evaluated: Condition A used 20 central tendency features, Condition B used 31 variability features, and Condition C used all 51 features combined. Hyperparameters were re-tuned independently for each condition using the same nested subject-wise validation structure described in Section 4.5.

Table 5.3 reports the results.

Condition	Features	Model	Acc	F1	PR-AUC	TN	FP	FN	TP
A: Central tendency	20	RF	0.672	0.802	0.770	0	33	10	88
A: Central tendency	20	XGB	0.725	0.839	0.771	0	33	3	95
B: Variability	31	RF	0.634	0.771	0.716	1	32	16	82
B: Variability	31	XGB	0.725	0.839	0.739	1	32	4	94
C: Combined	51	RF	0.680	0.807	0.708	1	32	10	88
C: Combined	51	XGB	0.718	0.834	0.692	1	32	5	93

Table 5.3: Feature ablation results across three conditions

The full 51-feature set did not improve classification over either feature group alone. Both XGBoost configurations using a single feature group matched or exceeded the combined set on F1, and central tendency features produced the strongest PR-AUC for both models, with substantial gains over the combined set (0.770 vs 0.708 for Random Forest and 0.771 vs 0.692 for XGBoost). Across all conditions, true negatives remained between 0 and 1, confirming that feature grouping did not resolve the control-class failure. Figure 5.1 visualises F1-score across conditions alongside the majority-class baseline.

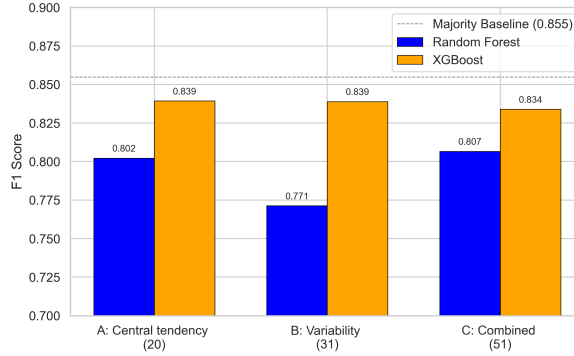


Figure 5.1: Feature Ablation: F1-Score by Condition and Model

5.2.3 Feature Importance and Stability

Table 5.4 reports the top ten features ranked by RF importance averaged across five outer folds from the primary model comparison on the full 51 feature set. XGBoost importances are shown alongside for comparison but are not independently ranked, as the purpose of the table is to highlight where the two models agree and disagree on the same features.

Rank	Feature	RF Importance	XGB Importance
1	Holdtime_skew	0.031	0.039
2	Flighttime_RL_mean	0.029	0.043
3	Holdtime_mean	0.027	0.027
4	Holdtime_RL_std	0.026	0.025
5	Flighttime_kurtosis	0.025	0.031
6	Holdtime_R_std	0.025	0.022
7	Holdtime_RL_mean	0.024	0.034
8	Flighttime_LR_mean	0.024	0.017
9	Holdtime_std	0.024	0.037
10	Holdtime_Iqr	0.024	0.032

Table 5.4: Top 10 features ranked by Random Forest importance in the primary 51-feature model

Feature importance was distributed relatively evenly across the 51 features for Random Forest, with the top-ranked feature (Holdtime_skew) contributing only 3.1% of total importance. XGBoost exhibited a similar spread but assigned different weights to some of the same features. Figure 5.2 visualises this comparison as a dot plot for the top 15 features. A notable disagreement within the top 10 was Holdtime_std, which Random

Forest ranked ninth (0.024), while XGBoost assigned it substantially higher importance (0.037). `Flighttime_RL_mean` was ranked second by Random Forest but received the highest importance from XGBoost (0.043). These disagreements suggest that the two learning strategies weight the same feature space differently, which motivates testing whether important features remain consistent across training samples.

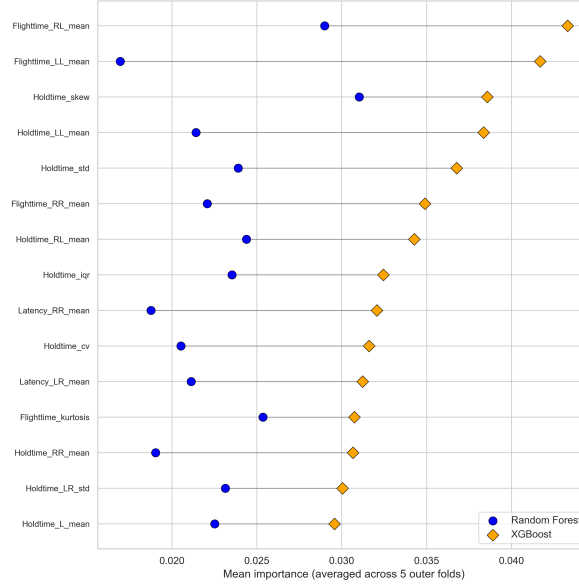


Figure 5.2: Feature importance comparison between Random Forest and XGBoost for the top 15 features

The feature stability analysis was restricted to the 20 central tendency features, based on the ablation finding that this group produced the strongest PR-AUC and provided the most defensible reduced representation. The stable subset was not selected globally from the full dataset. Instead, a fold-specific stable subset was identified inside each outer training fold using only that fold’s training data, following the procedure described in Section 4.7. Across the five outer folds, the union stable subset contained a mean of five features. The selected features varied by fold, indicating that importance rankings were not fully stable across different participant compositions. However, a small number of timing features recurred more frequently than the rest: `Flighttime_RL_mean` appeared in four of the five outer-fold subsets, while `Flighttime_LR_mean` and `Latency_LR_mean` each appeared in three.

Figure 5.3 summarises this recurrence pattern across the five outer folds. Only a small number of direction-specific timing features recurred, led by `Flighttime_RL_mean`, with `Flighttime_LR_mean` and `Latency_LR_mean` each appearing in three folds. This suggests a weak recurring signal rather than a single globally stable subset.

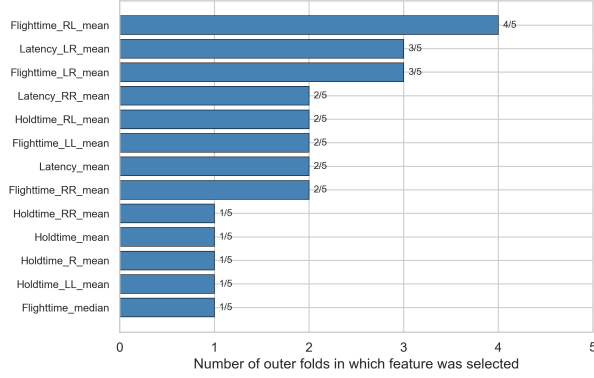


Figure 5.3: Frequency with which features appeared in the fold-specific stable subsets across the five outer folds

Figure 5.4 shows the model-specific stable features selected within each outer fold before the RF and XGBoost sets were combined by union for retraining. Blue cells indicate that a feature was selected as stable for that model in that outer fold. Stability was partial rather than absolute: no feature was selected by both models in every outer fold, and most selected features appeared only intermittently across the five splits. This supports the interpretation that the analysis identified recurring direction-specific timing features rather than a single fixed feature set.

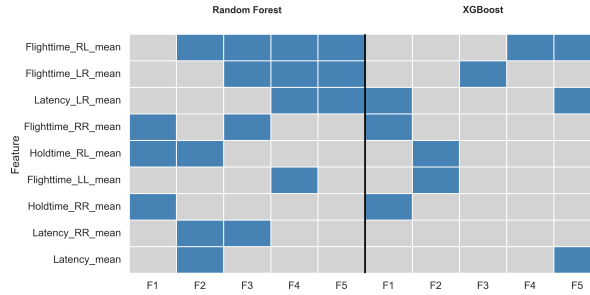


Figure 5.4: Feature stability heatmap showing model-specific stable feature selection across outer folds

5.2.4 Stable Feature Retraining

Stable-subset retraining used the fold-specific union stable subset identified inside each outer training fold. The subset was not fixed globally. It varied by fold, with a mean size of five features. Hyperparameters were re-tuned for the reduced feature space using the same nested subject-wise validation structure as the primary experiment. Table 5.5 compares the stable-subset results with the primary 51-feature configuration.

Configuration	Model	Acc	F1	PR-AUC	TN	FP	FN	TP
51 features	RF	0.680	0.807	0.708	1	32	10	88
Stable subset	RF	0.672	0.800	0.751	2	31	12	86
51 features	XGB	0.718	0.834	0.692	1	32	5	93
Stable subset	XGB	0.732	0.845	0.751	0	33	2	96

Table 5.5: Comparison of primary full-feature and fold-specific stable-subset retraining results

Stable-subset retraining did not produce broad performance gains over the full 51-feature configuration. Random Forest’s threshold metrics dropped slightly while PR-AUC improved from 0.708 to 0.751, and true negatives moved from one to two. XGBoost gained on accuracy, F1 and PR-AUC, but identified no control participant at all after retraining. The experiment therefore suggests dimensional efficiency and some ranking improvement, not a solution to the control-class failure.

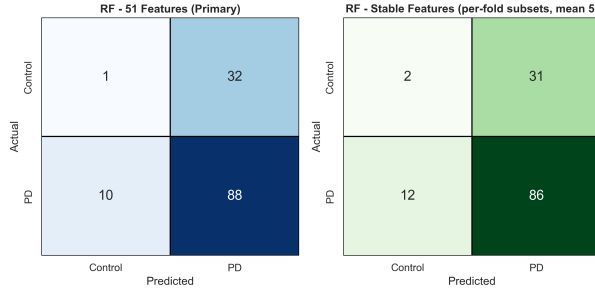


Figure 5.5: Random Forest confusion matrix comparison between full-feature and stable-subset configurations

Figure 5.5 focuses on Random Forest because it was the only model where stable-subset retraining increased correctly identified controls. The figure compares the full 51-feature configuration with the stable-subset configuration. The stable-subset model correctly identified two controls rather than one, but still misclassified 31 of 33 controls as Parkinson’s disease. The figure therefore shows a small shift away from total majority-class prediction, not a clinically adequate improvement in specificity.

5.2.5 SMOTE experiment

A supplementary experiment applied Synthetic Minority Over-sampling Technique (SMOTE) to test whether synthetic oversampling of the minority class altered classification behaviour. SMOTE was applied within each outer fold to the training set only, generating synthetic control instances until both classes were equal in size. The test set was not modified. Table 5.6 presents the results alongside the primary results without SMOTE.

Metric	RF (no SMOTE)	RF (SMOTE)	XGB (no SMOTE)	XGB (SMOTE)
Accuracy	0.680	0.520	0.718	0.550
Precision	0.736	0.683	0.746	0.723
Recall	0.904	0.655	0.954	0.657
F1	0.807	0.662	0.834	0.683
PR-AUC	0.708	0.708	0.692	0.695
TN	1	4	1	8
FP	32	29	32	25
FN	10	34	5	34
TP	88	64	93	64

Table 5.6: SMOTE comparison against primary 51-feature results

SMOTE reduced overall accuracy and F1 for both models, primarily through a large increase in false negatives. Both SMOTE configurations misclassified 34 PD participants

as controls, compared with 10 for Random Forest and 5 for XGBoost without SMOTE. True negatives rose from 1 to 4 for Random Forest and from 1 to 8 for XGBoost, showing that the decision boundary did move away from near-total Parkinson’s prediction. PR-AUC was effectively unchanged, indicating that SMOTE altered the default-threshold behaviour but did not materially improve ranking quality.

The failure of SMOTE to improve overall performance is likely attributable to the small control class. With only 33 real control participants, the synthetic examples were interpolated from a narrow region of feature space, producing variations of limited control patterns rather than genuinely new control behaviour. This shifted the decision boundary and increased control predictions, but at the cost of misclassifying many more PD participants. The results suggest that the limiting factor was not simply class imbalance but the limited diversity of the minority-class data.

5.2.6 Result Summary

Across all experiments, neither model provided strong evidence of reliable subject-wise discrimination between Parkinson’s and control participants. The ablation experiment identified central tendency as the stronger feature family, and the stability analysis showed that a small number of direction-specific timing features recurred across folds, but no fixed subset emerged. Stable-subset retraining improved ranking metrics and dimensional efficiency without resolving the control-class failure, and SMOTE shifted the decision boundary at a substantial cost to overall performance. The limiting factor was weak and unstable class-discriminative signal under subject-wise validation, not model complexity or feature volume.

5.3 Evaluation

5.3.1 Evaluation Against Objectives

The project met its three core objectives, but with a narrower empirical scope than originally proposed. Objective 1 required the acquisition, cleaning, and participant-level aggregation of the Tappy dataset, producing a 131-participant feature matrix with 51 engineered features. Objective 2 was achieved through the implementation and tuning of Random Forest and XGBoost under a consistent nested subject-wise validation framework. Objective 3 was also achieved: the models were evaluated and compared using standard classification metrics, and the analysis was extended through ablation, feature-stability, and SMOTE experiments, which strengthened the final evaluation beyond the original plan.

Meeting the objectives did not translate into clinically useful classification. The main outcome of the project was negative but still important: under subject-wise evaluation, neither model reliably separated Parkinson’s and control participants on the broader Tappy sample. The results directly answer the research question, because they show that the task becomes substantially harder once restrictive filtering is removed and evaluation is kept leakage-resistant.

5.3.2 Comparison with Prior Work

The contrast with Adams (2017) is analytically important rather than merely numerical. Where Adams’s near-perfect accuracy was obtained on a filtered 53-participant subset with LDA-projected features, the present project evaluated 131 participants on the engineered feature space directly. Combined with strict subject-wise validation and per-participant aggregation, the comparison is therefore less about raw accuracy and more about how reported performance changes when those experimental conditions are tightened.

Demir et al. (2023) reported accuracies between 81% and 100% using 378 features across 14 Tappy sub-datasets. Their results suggest that high performance is possible when feature volume, data partitioning, and ensemble design are expanded, but direct comparison remains limited because their evaluation procedure is not documented in enough detail to verify subject-wise separation. That matters because participant overlap between training and test data can inflate apparent performance in keystroke studies. In contrast, the present project prioritised transparency and leakage-resistant evaluation over maximising headline accuracy. The resulting performance was lower, but the conclusions are easier to interpret and more defensible as estimates of subject-wise generalisation.

The present findings are more consistent with the ecological validity argument in Arroyo-Gallego et al. (2018), who observed reduced performance when moving from supervised clinical collection to unsupervised home typing. Because the Tappy dataset was collected under unsupervised real-world conditions, weaker separation between Parkinson’s and control participants was expected. The results therefore suggest that modest performance observed here reflects the inherent difficulty of extracting robust disease signal from uncontrolled everyday typing, rather than a simple failure of chosen classifiers.

5.3.3 Limitations and Validity

Several limitations affect the conclusions that can be drawn from these results.

The most significant limitation is class imbalance. With 98 Parkinson’s and 33 controls, the dataset created a strong majority-class bias that both models largely followed. This was visible throughout the primary experiment, where both models identified only 1 of 33 controls despite high Parkinson’s recall. The SMOTE experiment demonstrated that rebalancing could increase control predictions, but only by also increasing false negatives. The problem was therefore not imbalance alone, but the limited amount of diversity of control-class signal available to the models.

A second limitation concerns the scope of the final modelling and analytical design. The pipeline used nested subject-wise validation, which is methodologically stronger than much of the prior Tappy literature, but it was still limited to two tree-based classifiers operating on participant-level summary features. This design improved interpretability and control over leakage, but it also removed within-participant temporal structure that may contain additional disease signal. The stability analysis was further restricted to the 20 central tendency features, on the basis of the ablation finding that this group carried the dominant signal, which means stability findings should be read as recurring patterns within the strongest feature family rather than universal predictors across all

51 features. The results should therefore be interpreted as evidence about what can be learned from aggregated keystroke features under leakage-resistant evaluation, not as a definitive upper bound for keystroke-based Parkinson’s detection.

A third limitation is the use of self-reported Parkinson’s disease in the Tappy dataset. Without independent clinical verification, some degree of label noise is unavoidable. That uncertainty places a ceiling on achievable performance and weakens the interpretation of borderline predictions, because some model error may reflect imperfect reference labels rather than purely model failure.

5.3.4 Interpretation of Key Findings

The research question asked how Random Forest and XGBoost behave when required to classify unseen Parkinson’s participants from keystroke dynamics, and what signal, if any, supports that classification. The findings provide a clear answer. Under subject-wise evaluation on the broader Tappy sample, neither model provided strong evidence of reliable discrimination between Parkinson’s and control participants. This does not imply that keystroke dynamics lacks clinical value as a biomarker. Rather, it suggests that the high accuracies reported by Adams (2017) were dependent on specific experimental conditions, particularly restrictive participant filtering and dimensionality reduction, that do not generalise when those conditions are removed.

The ablation experiment clarified where the discriminative signal was concentrated. Central tendency features consistently outperformed variability features and were at least as effective as the full 51-feature set on the strongest ranking-based metric. This suggests that average timing behaviour carried more stable class-related information than the variability measures engineered here. This does not imply that motor inconsistency is irrelevant. It indicates that, in this project, global variability summaries did not add enough complementary signal to offset the additional noise they introduced.

The stability analysis revealed that feature importance was only partially consistent across training samples. Fold-specific stable subsets varied across outer folds and no single feature remained stable across the entire analysis, although a small number of direction-specific central tendency features recurred more often than the rest, particularly `Flighttime_RL_mean`. The practical implication is that single-run importance rankings should not be treated as robust evidence of clinically meaningful predictors when participant composition changes between folds.

Stable-subset retraining did not overturn the main result, but it refined its interpretation. Restricting the models to fold-specific stable subsets improved some ranking metrics and reduced dimensionality, yet the control-class failure persisted. The stable-subset experiment therefore suggests that noise reduction through feature restriction was insufficient on its own, as the underlying class-discriminative signal remained too weak and unstable to support reliable separation, even in a reduced feature space.

5.4 Conclusion

The chapter has presented the primary subject-wise comparison of Random Forest and XGBoost on the broader Tappy sample, together with the ablation, stability, and SMOTE analyses that refined its interpretation. Across these experiments, the central pattern was consistent. Under leakage-resistant evaluation, the limiting factor was weak and unstable class-discriminative signal rather than model complexity or feature volume. The next two chapters consider the wider legal, ethical, and professional implications of that result and then draw together the project’s overall conclusions.

Chapter 6

Legal, Social, Ethical and Professional Issues

6.1 Introduction

This chapter examines the legal, social, ethical, and professional issues raised by the use of keystroke dynamics for Parkinson’s disease screening. The discussion is grounded in the results of this project rather than in assumptions of deployment readiness. Because the models showed near-zero specificity and did not provide clinically reliable discrimination between Parkinson’s and control participants, the chapter focuses primarily on the implications of handling behavioural health data and on the risk of misclassification and overclaiming model capability.

6.2 Research Data, Privacy, and Legal Compliance

The Tappy dataset used in this project is publicly available through PhysioNet (Goldberger et al. 2000) and complied with the dataset’s research-use terms through appropriate citation and academic use. The data was already anonymised before release, and the project did not collect any additional personal data or attempt to re-identify participants. In that sense, the legal and privacy risks within the scope of this academic study were limited.

However, the broader use of keystroke data for health-related inference raises more substantial concerns. Although keystroke timing does not record the content of what is typed, it is still behavioural data from which potentially sensitive health information may be inferred. In a real deployment context, such data would likely fall within the scope of health-related personal data under UK data protection law and would require clear consent, a lawful basis for processing, and strong limits on collection and use. The present project did not reach that stage, but the sensitivity of the underlying data remains an important consideration.

6.3 Ethical Risk of Misclassification

The most significant ethical issue raised by this project is the risk of misclassification. Both models correctly identified only 1 of 33 control participants, meaning most healthy

participants would be incorrectly classified as having Parkinson’s disease. In a screening context this would create a high false-positive burden, exposing users to unnecessary anxiety, further testing and avoidable clinical concern. Research on false-positive screening outcomes has shown such results can produce persistent psychological distress even when no disease is ultimately present (Bond et al. 2013). The risk is compounded by the appearance of high recall. A system that labels almost everyone as positive looks sensitive but provides little useful clinical information. The ethical concern is therefore not only model error in general, but the specific risk of presenting majority-class behaviour as meaningful disease detection.

A further ethical and practical concern is that impaired typing behaviour is not specific to PD. Slow or irregular keystroke patterns may also arise from fatigue, stress, intoxication, natural variation in typing skill, or other conditions affecting motor control. A system based only on keystroke timing therefore risks producing false positives not only because of model error, but because the measured behaviour itself is influenced by multiple non-Parkinsonian factors. This reinforces the conclusion that keystroke-based outputs should not be interpreted as diagnostic in isolation. A possible solution is that keystroke monitoring may be more useful in a longitudinal setting than in a single-session classification setting. Tracking sustained deviation from an individual’s own baseline over time could help distinguish persistent motor change from temporary effects. However, this approach was not tested in this project, and the results reported here do not support treating the present models as reliable stand-alone screening tools.

For that reason, the output of a system with the performance observed here should not be interpreted as diagnosis or even as a reliable stand-alone screening result. The main ethical lesson from the findings is that weak discrimination can still appear superficially persuasive if only headline scores are reported. Responsible evaluation therefore requires that confusion matrices, class-specific failures, and baseline comparisons are treated as central rather than secondary evidence.

6.4 Fairness, Accessibility and Representativeness

The project also raises questions of fairness and representativeness. After filtering, the dataset contained 98 Parkinson’s and 33 controls, producing a strong class imbalance that shaped model behaviour throughout the experiments. This class structure does not reflect real-world Parkinson’s prevalence and would make direct deployment particularly problematic, since false-positive behaviour could become even more severe when applied to a broader population.

Accessibility is a further limitation. Keystroke-based monitoring assumes that users type regularly on a physical keyboard, yet some of the people most relevant to Parkinson’s screening may type infrequently or may already experience motor impairment that limits keyboard use. A system based solely on keyboard interaction therefore risks excluding part of the population it is intended to assist. This does not make the approach worthless, but it limits its fairness and general applicability as a screening method.

6.5 Professional Responsibility and Honest Reporting

The main professional issue in this project was the responsibility to report results honestly and proportionately. The British Computer Society (BCS) principles of integrity, public interest, and avoidance of misleading claims are directly relevant here. Given the weakness of the observed control-class performance, it would have been misleading to present the models as clinically promising on the basis of accuracy or recall alone.

This makes transparency a professional obligation rather than a stylistic preference. The project therefore reported the majority-class baseline, confusion matrices, feature-stability limitations, and the failure of SMOTE to resolve the core problem. In this context, responsible conduct did not mean advancing a deployment argument, but clearly stating that the observed models were not adequate for dependable screening under the evaluation conditions used.

6.6 Conclusion

The legal, social, ethical, and professional issues in this project were shaped less by successful prediction than by the limits of models produced. The use of keystroke data for health inference raises genuine privacy and fairness concerns, but the most immediate issue identified here was the risk of unreliable classification being interpreted too strongly. A system with near-zero specificity would not be suitable as a stand-alone Parkinson’s screening tool. The most responsible conclusion was therefore not to argue for deployment, but to define the boundaries of what the results can and cannot do.

Chapter 7

Conclusion

7.1 Summary of Findings

This project asked whether Random Forest and XGBoost could reliably classify Parkinson’s disease from keystroke dynamics under nested subject-wise validation on a broader Tappy sample than used in prior work. The pipeline produced a 131-participant feature matrix with 51 engineered features and was extended through feature ablation, stability analysis, and a SMOTE experiment.

The central finding was that neither model reliably separated Parkinson’s from control participants. XGBoost was marginally ahead of Random Forest on threshold-dependent metrics, but both performed at or below the majority-class baseline and correctly identified only 1 of 33 controls across all folds. Under stricter evaluation conditions, the high accuracies reported in earlier Tappy work did not generalise.

The supplementary analyses refined this picture rather than overturning it. Central tendency features carried more usable signal than variability features. A small number of direction-specific timing measures recurred across folds without forming a fixed globally stable subset. Stable-subset retraining improved some ranking metrics and reduced dimensionality but did not resolve the control-class failure. SMOTE shifted the decision boundary toward more control predictions at substantial cost to overall performance. The limiting factor was weak and unstable class-discriminative signal under subject-wise validation, not model complexity or feature volume. All three project objectives were met, with Objective 3 extended beyond the original plan through the additional experiments.

7.2 Problems Encountered and Lessons Learnt

Several issues encountered during the project changed both its direction and its final contribution. The most significant was that the initial model comparison was not sufficient on its own to explain what the classifiers had learned. This led to the addition of the ablation experiment, which separated central tendency from variability features, and then to the feature stability analysis, which tested whether the most important features remained consistent across different training samples. These stages were not planned in the original proposal. They emerged through critical engagement with earlier results, and this reinforced the value of the CRISP-DM framework as an iterative process rather

than a rigid sequence.

A second lesson concerned the relationship between validation design and interpretation. Because the central research question depended on generalisation to unseen participants, the pipeline had to be held to a stricter standard than a simple train-test comparison or pre-tuned evaluation design. The final implementation therefore used nested subject-wise validation throughout. This became one of the most important methodological lessons of the project: when the contribution depends on the credibility of evaluation, validation design is not a technical detail but part of the research argument itself.

A third problem was class imbalance. With 98 Parkinson’s participants and only 33 controls, both models were drawn strongly towards the majority class. Neither class weighting nor SMOTE resolved this in a meaningful way. SMOTE increased the number of correctly identified controls, but it also introduced many more false negatives, showing that the issue was not simply the class ratio in abstract terms. The more fundamental limitation was the small and weakly defined control-class region in the available feature space. This demonstrated that algorithmic correction alone cannot compensate for limited minority-class diversity.

The project scope also changed substantially during development. The original proposal included multiple datasets and a broader range of models, including deep learning. After discussion with Dr Mohammad, it became clear that this breadth would likely produce a wide but shallow analysis with limited interpretive value. Narrowing the scope to a single dataset and a controlled comparison between two ensemble classifiers produced a stronger dissertation because it made deeper analysis possible. The narrower scope created room for the ablation, stability, and SMOTE experiments, which contributed more to the final argument than a broader but less focused comparison would have done.

The main lesson from the project was therefore methodological. In this domain, strong reported accuracy is less important than whether the evaluation design, feature representation, and interpretation support defensible conclusions about unseen participants. The project did not produce a clinically useful classifier, but it did produce a clearer account of what cannot be learned from the Tappy dataset under stricter evaluation conditions.

7.3 Future Work

The clearest direction for future work is the use of larger and more balanced datasets. The principal limitation of the Tappy dataset in this study was the small number of control participants combined with self-reported diagnosis labels. A dataset with reduced class imbalance, clinically verified diagnoses, and more standardised collection conditions would provide a firmer basis for evaluating classifier performance. A particularly useful resource would combine the ecological realism of Tappy with the diagnostic reliability of more clinically controlled datasets such as NeuroQWERTY.

A second direction is the investigation of temporal feature representation. In the present project, all keystrokes for a participant were aggregated into a single feature vector, which removed information about how typing behaviour changed over time. Since Parkinson’s

disease is progressive and may affect movement inconsistently, a session level or longitudinal representation may capture signal that participant-level aggregation suppresses. Any such design would, however, require careful grouping controls to prevent within-subject leakage.

A third direction follows directly from the stability analysis. The features that recurred most across fold-specific stable subsets, including `Flighttime_RL_mean`, `Flighttime_LR_mean` and `Latency_LR_mean`, capture asymmetric transition timing that is clinically plausible in Parkinson’s disease. Future work could examine whether these measures vary systematically with handedness, disease stage or medication status. More broadly, the stable-subset experiment suggests that feature restriction based on recurring fold-wise importance may be more informative on this problem than simply increasing feature volume. Testing that idea on other keystroke or behavioural biomarker datasets would help determine whether the pattern observed here is specific to Tappy or reflects a more general property of small, noisy biomedical typing datasets.

7.4 Concluding Remarks

The contribution of this project is not the identification of a high-performing classifier on the Tappy dataset, but a transparent, leakage-resistant benchmark that shows where the high accuracies reported in earlier work do and do not generalise. Under broader subject-wise evaluation, central tendency features carried more usable signal than variability features, and a small set of direction-specific timing measures recurred across folds. However, the available signal remained too weak and unstable to support reliable separation between Parkinson’s and control participants. That bounded result, rather than any single accuracy figure, is what the dissertation contributes.

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Appendix A

Appendix A – Project Proposal

The original proposal planned a multi-dataset comparison across Tappy, NeuroQWERTY, and Mendeley using Random Forest, XGBoost, and a transformer-based deep learning model. After speaking to Dr Mohammad, the scope was narrowed to a single dataset (Tappy) for a clearer motivation with a controlled pairwise comparison between Random Forest and XGBoost. This revision enabled deeper analysis through feature ablation, stability analysis, and a SMOTE experiment, which would not have been feasible under the original breadth. The original proposal is included below.

UNIVERSITY OF GREENWICH
FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF COMPUTING AND MATHEMATICAL SCIENCES

COMP1682 Final Year Project Proposal

Keystroke dynamics to detect Parkinson's disease

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001302840

BSc (Hons) Computer Science (with artificial intelligence)

Supervisor: Al-Rifaie, Mohammad Majid

Word count: 1740

November 2025

DECLARATION OF AI USE

Please complete this part when you have used AI during the process of undertaking this assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources: – YES
 1. Detection of Parkinson's disease with keystroke data
 2. Application of artificial intelligence in community-based primary health care
 3. Diagnosing Parkinson's disease via behavioural biometrics
 4. Detecting motor impairment in early Parkinson's disease via digitized NeuroQWERTY index
 5. High-accuracy detection of early Parkinson's disease using multiple characteristics of finger movement while typing

- In other ways, as described below: – YES

After the document was completed, AI checked if I included all the template had asked for and checked tone and flow.

SYNOPSIS

Parkinson's disease (PD) is a progressive neurodegenerative disorder that primarily affects motor function. As there is no single test for PD, diagnosis relies mainly on patient history and neurological examinations. Procedures such as DaTscan or a skin biopsy may assist however, these tests are often invasive, costly, and not always conclusive. These diagnostics require patients to be examined by a doctor and for results to be returned, which can mean that the patient already experiences significant motor neuron loss.

Recent research suggests that subtle motor irregularities, such as tremors and reduced fine-motor control, can emerge long before clinical symptoms are obvious. This project investigates whether these minor changes in fine motor control can be detected through analysis of typing behaviour and keystroke dynamics, ultimately providing a non-invasive digital biomarker for early PD detection.

The project will utilise publicly available keystroke datasets, including Tappy, NQMITCSxPD, and Mendeley, to train, evaluate, and compare machine learning models and deep learning models, such as Extreme Gradient Boosting (XGBoost), Random Forest (RF), and Transformer-based models. Each model will be evaluated on its ability to classify Parkinsonian and control participants using standard metrics such as accuracy, precision, recall, harmonic mean (F1-score), and Receiver Operating Characteristics – Area Under the Curve (ROC AUC). This evaluation will be supported by feature importance analysis to interpret model behaviour.

Keywords: Parkinson's disease, keystroke dynamics, digital biomarkers, fine motor control, machine learning, deep learning, temporal typing patterns, Random Forest, XGBoost, Transformer, F1-score, ROC AUC, early detection

1. AIM AND OBJECTIVES

1.1 Aim

This project aims to develop, evaluate and compare machine learning models to identify the most effective method for distinguishing Parkinson's disease from control participants using keystroke dynamics.

1.2 Objectives

1.2.1 Objective 1 – Dataset normalisation and preprocessing

- 1.2.1.1 Find, download and consolidate publicly available keystroke datasets examples include Tappy and Mendeley [14 days]
- 1.2.1.2 Clean and standardise data formats, handle missing or inconsistent entries [28 day]
- 1.2.1.3 Perform and document data analysis highlighting distributions, correlations and class balance [21 days]
- 1.2.1.4 Deliverable: A unified dataset which is documented ready for model training with summary plots and feature statistics.

1.2.2 Objective 2 – Model development and optimisation

- 1.2.2.1 Implement and train multiple machine-learning algorithms under consistent preprocessing conditions [35 days]
- 1.2.2.2 Optimise model parameters to ensure performance remains robust and applicable [21 days]
- 1.2.2.3 Deliverable: Two fully trained and optimised models saved for evaluation, with completed documentation of all parameters used and training logs.

1.2.3 Objective 3 – Model evaluation and Reporting

- 1.2.3.1 Evaluate model performance using the following defined metrics: accuracy, precision, recall, F1-score and Receiver Operating Characteristic – Area under the Curve (ROC AUC) [21 days]
- 1.2.3.2 Compare and document results to identify the most effective classification approach. [21 days]
- 1.2.3.3 Produce a comprehensive evaluation report summarising documentation of Objective 1, 2 and all evaluation metrics and findings [21 days]
- 1.2.3.4 Deliverable: Final project report including comparative analysis, performance tables and discussion of results.

2. BACKGROUND RESEARCH AND PROJECT RATIONALE

Parkinson's disease (PD) is one of the most prevalent neurodegenerative disorders, affecting almost ten million people worldwide. Motor symptoms only become clinically evident after substantial neuron loss, making early detection crucial. Traditional diagnosis relies on clinical observation, which is subjective and often delayed, while confirmatory tests such as DaTscan remain invasive and costly. This provides a strong motivation to identify objective, non-invasive, and accessible digital biomarkers that can detect Parkinson's disease earlier in its progression.

Other non-invasive approaches for early PD detection include handwriting analysis, speech pattern recognition, gait tracking, and facial expression monitoring. Although these methods have produced promising results, but they often require specialised sensors, controlled environments, or, in some cases, high-fidelity audio and video data. These requirements limit scalability and raise legal, social, ethical, and professional (LSEPi) concerns.

Tappy, NeuroQWERTY and Mendeley are three open-source datasets that provide structured and natural typing data with clinically verified labels and diverse participant groups.

Adams et al. (2017) achieved a high accuracy in distinguishing the early-stage PD patients from healthy control patients using multiple typing pattern features extracted from the Tappy dataset. Their work validated the feasibility of keystroke-based biomarkers but relied on handcrafted statistical features and a small number of participants, which negatively impacted the scalability potential. Similarly, Arroyo-Gallego et al. (2017) introduced the NeuroQWERTY Index, which was derived from the NQMITCSxPD dataset, showing that fine-grained key-hold and flight time variability correlated with motor impairment severity. The most significant weakness of Arroyo's study was that, as it was a controlled experimental setting, it limited how well it can be generalised to the real world, and the proposed index was tested on a single dataset without independent replication.

More recent approaches have explored deep-learning models to capture non-linear temporal dependencies in keystroke data. Ruiz et al (2022) applied a convolutional neural network (CNN) to at-home typing data, reporting improved feature extraction and classification accuracy when compared to traditional machine learning baselines. Like the other studies, it lacked subject-level validation and interpretability analysis. Ruiz later addressed part of this issue in 2025 through a cross-dataset time series model using a combination of Tappy and NeuroQWERTY data, but generalisation results varied considerably between participants, highlighting the underlying difficulty of a truly consistent benchmark.

Across the publicly available studies that have been reviewed above, the main methodological limitations to prior work are as follows:

1. Data leakage and inflated accuracy arising from record-level rather than subject-wise evaluations
2. Over-reliance on handcrafted features that failed to model temporal motor fluctuations.
3. Limited interpretability and cross-dataset testing, hindering reproducibility and therefore real-world deployment

While keystroke dynamics shows strong potential for early detection, variation in individual typing behaviour remains a challenge. Factors such as natural typing speeds, fatigue, stress, or intoxication can temporarily alter motor control and affect model predictions. Accounting for this variability is crucial to prevent false positives and ensure that predictions reflect sustained neurological patterns rather than short-term behavioural changes. Despite these challenges, keystroke dynamics offers clear

advantages, including its ability to be deployed globally, independent of language or typing context, making it an inclusive and scalable approach for digital health monitoring.

3. METHODOLOGY AND TECHNICAL APPROACH

This project adopts an experimental, comparative machine learning workflow to evaluate different classification models for detecting Parkinson's disease (PD) from keystroke timing data. The study will involve training and testing multiple algorithms under identical preprocessing and validation conditions to ensure fair comparison and reproducibility.

The primary objective is to determine which model architecture provides the best balance between accuracy, interpretability and generalisation across participants and datasets. All data used for training and testing will be open source and anonymised to ensure ethical compliance. Model performance will be evaluated using standard classification metrics, and every stage of the pipeline will be documented to ensure full transparency.

Early work relied on handcrafted statistical features such as key hold duration, flight time and latency variance, paired with classical models like Support Vector Machines and Random Forests. These studies achieved strong dataset accuracy but often suffered from data leakage; participant data was found in both training and testing sets, artificially inflating accuracy and reducing generalisability. Examples of these are Adams et al. (2017) and Arroyo-Gallego et al. (2017).

More recent research introduced a convolutional and sequence-based deep learning architecture capable of modelling non-linear temporal dependencies in typing data, meaning the models can learn how the timing between key presses changes over time, rather than treating each key event separately. This approach led to measurable improvements in feature representation and classification as demonstrated in Ruiz et al.'s (2022) study.

Building on the methodologies adopted in previous studies, this project will apply the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, an established and widely used workflow for data-driven experimentation. CRISP-DM provides a structured workflow ensuring consistency and traceability during model development. The framework will be divided into five separate stages: data understanding, data preparation, modelling, evaluation and reporting. CRISP-DM was selected over other approaches because it offers a clear structure for documentation, traceability for decisions such as tuning hyperparameters and the integration of multiple datasets.

Data preparation will focus on cleaning and standardising the Tappy, NeuroQWERTY, and Mendeley datasets. To create a consistent and reliable experimental base, each dataset will be downloaded from its respective public repository.

Data cleaning will involve removing any corrupted or duplicated records, handling any missing timestamps, and normalising time measures to uniform the scale in milliseconds. Using metadata is key to maintaining separation during training and testing, therefore preventing data leakage.

Feature engineering will calculate common keystroke metrics such as hold time, flight time, and inter-key latency. This will allow statistical descriptors such as the mean and standard deviation for each typing session. Before calculating these statistical descriptors, all numerical features are standardised using z-score normalisation, which removes scale bias.

Model development will focus on implementing and comparing three supervised algorithms: Random Forests, Extreme Gradient Boosting (XGBoost) and a transformer-based deep learning architecture. The takeaway from this project is to also see if a transformer model can allow deep architecture to capture complex motor patterns that handcrafted features may overlook.

All implementations will be written in Python using Scikit-learn, PyTorch, and XGBoost, with all models sharing an identical preprocessing pipeline. Model outputs will later be assessed through standard performance metrics to identify the best-performing architecture.

4. EVALUATION PLAN

Once training and testing are completed, model performance will be measured using industry standard metrics, accuracy, precision, F1-score, and ROC AUC are all examples of these. Indicators will aid in assessing how well each individual model was able to differentiate between a participant with PD and a control participant. The project will also include cross-data analysis to showcase how the models adapt to new writing styles. Combining these methods will hopefully create a reproducible benchmark that will aid in being a digital biomarker for early detection in PD through keystroke dynamics.

5. LSEPI CONSIDERATIONS

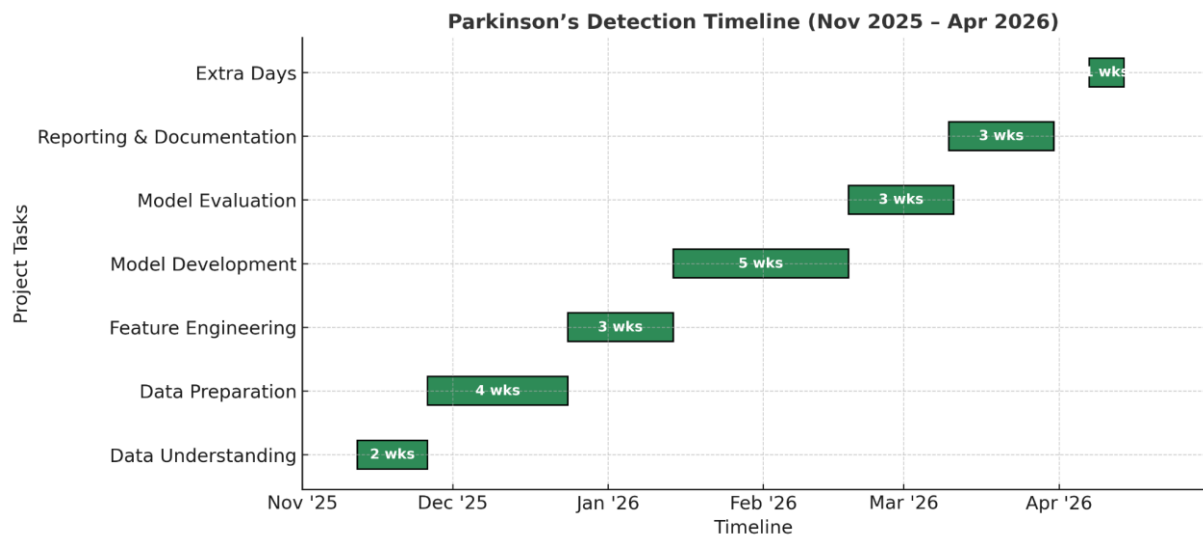
All datasets used in this project are open source and publicly available and have already been anonymised, meaning that no personally identifiable information is processed. From a legal and ethical viewpoint, all data handling will follow research principles and will follow appropriate citation and acknowledgement of data sources. Socially, the project positively promotes inclusivity by exploring accessible, low-cost digital biomarkers that could support early Parkinson's detection regardless of spoken language. All artificial intelligence (AI) models will be evaluated responsibly, avoiding claims

beyond the scope of academic experimentation, which aligns with the British Computer Society (BCS) code of conduct by prioritising transparency, reproducibility and public benefit in AI research.

6. RISK ASSESSMENT

Risk	Likelihood	Impact	Mitigation measures
Data unavailability	Low	High	Using multiple data sources to avoid dependency on a single type of source.
Poor data quality	Low	High	Apply Preprocessing techniques such as normalisation and deduplication.
Time constraints	Medium	High	Develop a detailed plan outlining each project stage and allocate time for potential delays or technical issues.
LSEPi compliance	Low	High	Ensure all data handling and citations adhere to Greenwich's ethical guidelines and GDPR standards.
Model performance issues	Medium	High	Maintain a clear understanding of model parameters and use validation techniques such as cross-validation and hyperparameter tuning to optimise performance.

7. PROJECT PLAN AND TIMELINE



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